# Kvist & Bjælke ApS Network plan

## 2.1 The company network

ernvervsakaaemi

### 2.1 The Company's network

Kvist & Bjælke ApS has several employees and IT systems that must operate within the company's network.

The company currently has the following devices:

| Device | Quantity | Description |
|---|---|---|
| Desktop computers | 15 | Workstations used for administration and office tasks |
| Laptops | 10 | Used by employees working from different locations |
| Network printers | 3 | Shared printers available to employees |
| Servers | 3 | Hosting internal systems and applications |
| IP phones | 12 | Telephony using the company's network |
| Wireless access points | 4 | Wireless network access for employees and guests |
| Surveillance cameras | 6 | IP-based video surveillance |
| Network equipment | As required | Switches, routers, and other necessary network equipment |
| Guest devices | Variable | Smartphones and computers belonging to visitors |

## 2.1.1 Development of an Ip-adressing plan

| ID | Name | Devices | Subnet | Gateway | DHCP |
|---|---|---|---|---|---|
| 70 | Office VLAN 1 | Desktop Computers: 15<br>Laptops : 10<br>Total devices: 25 | 10.5.70.2/26 | 10.5.70.2 | 10.5.70.2<br>2 to .70.62 |
| 73 | Office VLAN 2 | Network printers servers :3<br>Total devices: 3 | 10.5.73.2/29 | 10.5.73.2 | 10.5.73.2<br>2 to .73.62 |

| | | | | | |
|---|---|---|---|---|---|
| 72 | Office VLAN 3 | Surveillance cameras Network equipment: 6<br>Total devices: 6 | 10.5.72.2/28 | 10.5.72.2 | None(Static) |
| 71 | Office VLAN 4 | Servers: 3 | 10.5.71.2/29 | 10.5.71.2 | None(Static) |
| 20 | Phone VLAN 1 | Ip Phones: 12 | 10.5.20.2/27 | 10.5.20.2 | 10.5.20.2<br>2 to .20.32 |
| 30 | Lobby VLAN 1 | Guest devices | 10.5.30.2/26 | 10.5.30.2 | 10.5.30.2<br>2 to .30.62 |
| 31 | Lobby VLAN 2 | Employees | 10.5.31.2/29 | 10.5.31.2 | 10.5.31.2<br>2 to .31.6 |

## Subnet decisions

The reason we have chosen the specific subnet masks, is that there could be hired some new employees soon. The other reason is that we expect the company to expand in the next few years, so that the company will have a buffer for the future. That would also make so we don't have to touch the network configuration in the near future.

## Network devices

We have decided that the company needs one firewall for the network. The firewall is setup so it only allows data that follows its configuration.  The router is a part of the firewall so it's a firewall/router combo.

We will have four access points in total. We will be having two in the office, one in the lobby and one meetings room.

The amount of switches we want is around 48 port unit cause a 24 port unit would be too close to the bare minimum.

## 2.1.2 Subnetting

Office VLAN 4

IP address: 10.5.71.2/29

Office VLAN 4 has three servers on it so we have given it /29 that allows the subnet to have up to 6 devices on it. We have given it that cause /30 only allows 4 devices and /28 allows 16 devices. 6 devices is twice the amount of servers the company have at the moment which allows them to expand the company without worrying about space on the subnet.

The reason why the servers have their own subnet is cause it makes it incased from the rest of the network.

Office VLAN 2

IP address: 10.5.73.2/29

Office VLAN 2 has three printers connected to it. We have made many of the same decisions as with the Office VLAN 4. One of the key difference is that we made the printers incased cause printers are usually more unsafe then other devices.